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RESEARCH

Facile 3-chloro-1*H*-indazole based 1,3,4-thiadiazole derivatives as novel thymidine phosphorylase and anti-diabetic inhibitors: Experimental, in-vitro and molecular modelling approaches

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Abstract
A series of novel 3-chloro-1H-indazole-based 1,3,4-thiadiazole derivatives (**1–15**) were synthesized, characterized and evaluated for their inhibitory activity against thymidine phosphorylase and α-glucosidase. Several compounds showed potent dual inhibition, with compound **4** exhibiting the highest activity ($IC_{50} = 4.70 \pm 1.34$ μM for thymidine phosphorylase and $IC_{50} = 1.38 \pm 0.46$ μM for α-glucosidase), surpassing the standard inhibitors 7-deazaxanthine ($IC_{50} = 12.42 \pm 1.27$ μM) and acarbose ($IC_{50} = 5.97 \pm 0.35$ μM). Moreover, other synthesized derivatives included compounds 9, 2, 1 and 6 all showing improved inhibition profiles. Structure activity relationship analysis revealed that the nature and position of substituents on the phenyl ring play a critical role in modulating the enzyme inhibition potential. Similarly, molecular docking confirmed favorable binding interactions with the active sites of both enzymes, while ADMET profiling indicated good drug-likeness and pharmacokinetic potential. These findings suggest that thiadiazole-indazole hybrids represent promising lead candidates for the development of dual targets enzymes inhibitors with potential therapeutic applications in metabolic and proliferative disorders.

Keywords Indazole · 1,3,4-thiadiazole · Thymidine phosphorylase · Anti-diabetic · Molecular modelling

Introduction

Enzyme thymidine phosphorylase (TP), utilized in restoration pathway of pyrimidine was discovered in 1954. The main function of TP is recovering the pyrimidine nucleosides. These enzymes were generated during degradation of RNA or DNA. TP actually converts thymidine and 20-deoxyuridine into 2-a-D-deoxyribose-1-phosphate (2DDR-1P) and respective bases (Friedkin and Roberts 1954; Iltzsch et al. 1985). Despite being reversed the main function of TP remains the same, i.e., proteolytic activity. In the process of deoxyribosyl transferase activity, the deoxyribosyl part of TP gets transferred from one pyrimidine base to another (Schwartz 1971). The results is the confirming of the formation of new pyrimidine molecules (Desgranges et al. 1983; Nemr et al. 2019). A range of pyrimidines or pyridine nucleosides with antiviral or anti-cancer potency are consumed by TP. For example, 5-fluorouracil (5FU), 5-trifluorothymidine (TFT), 5-(E)-(2-bromovinyl)−20-deoxyuridine (BVDU) are few examples of pyrimidines or pyridines nucleosides containing basic structures (El-Zoghbi et al. 2023; Nemr and Aboulmagd 2020; Fadaly et al. 2023; Nemr et al. 2023).

Diabetes mellitus (DM) is a chronic disorder of metabolism (Kasturi et al. 2017). It is characterized by either decreased secretion of insulin or ineffective insulin production due to organ's failure (Yue et al. 2017; Shah et al. 2018; Olatunji et al. 2017). The non-uniform secretion of insulin causes an increase in blood glucose levels, commonly termed hyperglycemia (Salar et al. 2017). The worldwide cases effected by DM are gradually elevating. The pancreas

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secretes amylase (EC. 3.2.1.1), a hydrolyzing enzyme which is responsible for conversion of starch into glycogen and maltose. This maltose gets further converted into glucose in small intestine by glucosidase enzyme (Sun et al. 2017). The conversion of starch into maltose and then glucose results in elevated blood glucose level. It is typically termed as post-prandial glycemia (PPHG). PPHG is the foremost measure of type-II DM (Campos 2012). The elevated glucose levels in blood lead to vascular disorders like retinopathy, neuropathy and nephropathy (Rask-Madsen and King 2013; Miller et al. 2014). Different therapeutic techniques involve the reduction in the enzymatic activity of pancreatic and intestinal glucosidase are employed in the treatment of DM. Miglitol, voglibose, and acarbose were few drugs commercially available for the treatment of type-II DM. But the problem with the use of these drugs is their side effects like meteorism, diarrhea, and stomach discomfort which arises the needs of new drugs (Raghu et al. 2019; Gandham et al. 2024).

Indazole molecule is also known as benzopyrazole or isoindazolone belonging to heterocyclic compounds with two nitrogen atoms in the structure (Rayawgol et al. 2024). The presence of ten pi-electrons makes indazole behave identical to other nitrogen containing heterocycles like pyrazole, pyrrole and pyridine (Bastos et al. 2024b). Indazoles have been used as a framework in the synthesis of a variety of biological and medical components (He et al. 2024). Therefore, researchers have developed a wide range of approaches globally to prepare these heterocycles. Indazole has been utilized in various biological, agricultural, and industrial applications (Bastos et al. 2024a).

In this paper novel 3-chloro-1*H*-indazole based 1,3,4-thiadiazole derivatives (1–15) derivatives are synthesized and their enhanced inhibitory potential against thymidine phosphorylase and α-glucosidase were evaluated with comparison to standard drugs. The protein–ligand binding interactions of synthesized derivatives are investigated by employing molecular docking studies. The lead compound of the synthesized derivatives was analyzed to investigate the drug-like properties. The rationalization of the synthesized compounds was compared with the reported one were compared it with [25] (Khan et al. 2023c), [26] (Ullah et al. 2021) and [27] (Taha and Wadood 2018) respectively as follows in Fig. 1, Khan et al., 2024e.

Experimental

Material and methods

Sigma Aldrich, USA was approached for the purchase of chemicals. For NMR spectroscopy, Avance Bruker Machine was employed. To record high resolution electron impact

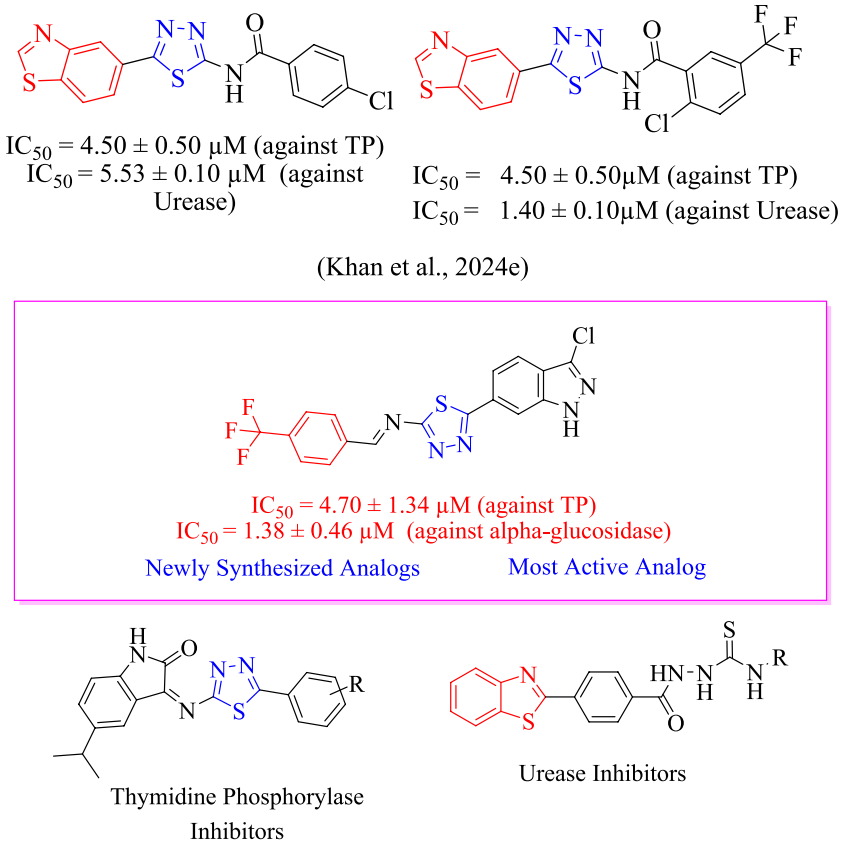


Fig. 1 Rationalization for the current study

mass spectra (HR-EIMS), Finnigan MAT-311A (Germany) mass spectrometer was used. Pre-coated silica gel aluminum plates (Kieselgel 60, 254, E. Merck, Germany), was employed for the thin layer chromatography (TLC). UV spectrophotometry was conducted at a wavelength of 254 nm and 365 nm to visualize the TLC chromatograms.

Synthesis of 3-chloro-1H-indazole-based thiosemicarbazone intermediate (II)

The first step consists of mixing of 3-chloro-1*H*-indazole-5-carbaldehyde **(I)** with thiosemicarbazide in ethanol solvent in the presence of few drops of acetic acid. The system was refluxed for six hours to obtain the intermediate moiety 3-chloro-1*H*-indazole-based thiosemicarbazone **(II)**.

Synthesis of 3-chloro-1H-indazole-based 2-amino-thidaizole substrate (III)

In ethanol as solvent, the intermediate (II) reacted with molecular iodine leads to oxidative cyclization followed by addition of potassium carbonate in 1,4-dioxane drop wisely in reaction mixture and stirring for 13 to 14 h to yield 3-chloro-1*H*-indazole-based 2-amino-thidaizole substrate **(III)** intermediate.

Synthesis of 3-chloro-1H-indazole based 1,3,4-thiadiazole derivatives (1–15)

This intermediate **(III)** was dissolved in ethanol, following addition of triethylamine as base and differently substituted aldehydes. After the completion of reaction, the conversion of intermediate **(III)** into 3-chloro-1*H*-indazole based 1,3,4-thiadiazole derivatives (**1–15**) in appreciable yields.

Spectral Analysis

(E)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)−1-(4-nitrophenyl)methanimine (1)

Yield: 60%, state = yellow solid, Melting Point = 192–193 °C; FTIR (KBr): (cm^{-1}), 3420 (NH), 3217 (Ar–CH), 1632 (C = N), 1405 (C = C); ^{1}H-NMR (600 MHz, DMSO-d6): *δ* 11.64 (s, 1H, -NH), 9.41 (s, 1H, imine-H), 8.81 (d, 2H, *J* = 7.2 Hz, Ar–H), 8.33 (d, 2H, *J* = 8.2 Hz, Ar–H), 8.07 (d, 1H, *J* = 7.2 Hz, Ar–H), 7.43 (s, 1H, Ar–H), 7.18 (d, 1H, *J* = 7.4 Hz, Ar–H); ^{13}C-NMR (150 MHz, DMSO-d6): *δ* 175.1, 172.6, 164.8, 164.3, 163.3, 161.4, 147.6, 143.1, 136.6, 135.8, 131.8, 127.6, 125.3, 124.6, 122.1; HREI-MS: m/z calcld for $C_{16}H_{19}ClN_6O_2S$, $[M]^+$ 384.1642 Found: 384.1392.

(E)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)−1-(2,4-dichlorophenyl)methanimine (2)

Yield: 72%, state = yellow solid, Melting Point = 198–200 °C; FTIR (KBr): (cm^{-1}), 3416 (NH), 3245 (Ar–CH), 1630 (C = N), 1432 (C = C); ^{1}H-NMR (600 MHz, DMSO-d6): δ 11.32 (s, 1H, -NH), 9.72 (s, 1H, imine-H), 8.79 (d, 1H, *J* = 8.4 Hz, Ar–H), 8.08 (d, 1H, *J* = 8.2 Hz, Ar–H), 7.50 (s, 1H, Ar–H), 7.34 (s, 1H, Ar–H), 7.13 (d, 1H, *J* = 6.9 Hz, Ar–H), 6.68 (d, 1H, *J* = 7.0 Hz, Ar–H); ^{13}C-NMR (150 MHz, DMSO-d6): *δ* 177.6, 173.5, 166.8, 163.6, 162.6, 151.9, 143.6, 141.5, 139.2, 135.5, 135.3, 125.9, 124.6, 119.1; HREI-MS: m/z calcld for $C_{16}H_8Cl_3N_5S$, $[M]^+$ 408.0748 Found: 408.0655.

(E)−4-(((5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)imino)methyl)phenol (3)

Yield: 61%, state = off-white solid, Melting Point = 161–163 °C; FTIR (KBr): (cm^{-1}), 3472 (NH), 3243 (Ar–CH), 1605 (C = N), 1394 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6): δ* 11.30 (s, 1H, -NH), 10.39 (s, 1H, -OH), 9.43 (s, 1H, imine-H), 8.37 (d, 1H, *J* = 7.4 Hz, Ar–H), 8.18 (s, 1H, Ar–H), 7.43 (d, 2H, *J* = 8.1 Hz, Ar–H), 7.29 (d, 1H, *J* = 7.2 Hz, Ar–H), 6.62 (d, 2H, *J* = 7.4 Hz, Ar–H); ^{13}C-NMR (150 MHz, DMSO-*d6): δ* 176.1, 174.4, 164.7, 162.5, 161.7, 153.0, 152.5, 144.5, 135.8, 132.8, 132.7, 123.7, 120.7, 112.5; HREI-MS: m/z calcld for $C_{16}H_{10}ClN_5OS$, $[M]^+$ 355.4773 Found: 355.4752.

(E)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)−1-(4-(trifluoromethyl)phenyl)-methanimine (4)

Yield: 60%, state = green solid, Melting Point = 156–157 °C; FTIR (KBr): (cm^{-1}), 3470 (NH), 3243 (Ar–CH), 1608 (C = N), 1408 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6): δ* 11.32 (s, 1H, -NH), 9.74 (s, 1H, imine-H), 8.88 (d, 1H, *J* = 7.1 Hz, Ar–H), 8.58 (d, 2H, *J* = 8.3 Hz, Ar–H), 7.88 (s, 1H, Ar–H), 7.53 (d, 1H, *J* = 7.4 Hz, Ar–H), 6.88 (d, 2H, *J* = 7.8 Hz, Ar–H); 13C-NMR (150 MHz, DMSO-d6): δ 176.3, 169.2, 167.6, 166.5, 156.5, 155.8, 149.5, 138.1, 133.8, 133.2, 128.8, 123.5, 120.5, 113.1, 108.8; HREI-MS: m/z calcld for $C_{17}H_9ClF_3N_5S$, $[M]^+$ 407.8354 Found: 407.8152.

(E)−3-(((5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)imino)methyl)−5-methylphenol (5)

Yield: 64%, state = off-white solid, Melting Point = 159–160 °C; FTIR (KBr): (cm^{-1}), 3484 (NH), 3261 (Ar–CH), 2929 (sp^3-CH), 1620 (C = N), 1410 (C = C); ^{1}H-NMR (600 MHz, DMSO-d6): δ 11.45 (s, 1H, -NH), 10.28 (s, 1H, Ar-OH), 9.17 (s, 1H, imine-H), 8.53 (d, 1H, *J* = 7.4 Hz, Ar–H), 8.24 (s, 1H, Ar–H), 8.02 (d, 1H, *J* = 7.8 Hz, Ar–H), 7.81 (s, 1H,

Ar–H), 7.64 (s, 1H, Ar–H); ^{13}C-NMR (150 MHz, DMSO-d6): δ 173.7, 168.7, 158.5, 144.7, 143.1, 133.8, 130.5, 130.3, 129.1, 126.5, 123.8, 120.8, 120.3, 117.3, 109.0, 28.9; HREI-MS: m/z calcld for $C_{17}H_{12}ClN_5OS$, $[M]^+$ 369.0491 Found: 369.0466.

(E)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)−1-(2-fluoro-4-methylphenyl)methanimine (6)

Yield: 73%, state = green solid, Melting Point = 163–164 °C; FTIR (KBr): (cm^{-1}), 3471 (NH), 3245 (Ar–CH), 2954 (sp^3-CH), 1619 (C = N), 1493 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6*): δ 11.50 (s, 1H, -NH), 9.43 (s, 1H, imine-H), 8.50 (d, 1H, J = 8.2 Hz, Ar–H), 8.32 (d, 1H, J = 7.3 Hz, Ar–H), 8.16 (s, 1H, Ar–H), 8.09 (d, 1H, J = 8.0 Hz, Ar–H), 7.94 (d, 1H, J = 7.4 Hz, Ar–H), 2.94 (s, 3H, Ar-CH_3); ^{13}C-NMR (150 MHz, DMSO-*d6*): δ 174.8, 163.9, 150.5, 152.7, 149.2, 146.3, 144.0, 143.6, 143.4, 140.3, 140.9, 136.7, 132.9, 132.6, 128.1, 124.7, 122.9, 26.9; HREI-MS: m/z calcld for $C_{17}H_{11}ClFN_5S$, $[M]^+$ 371.5821 Found: 371.5810.

(E)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)−1-(3-methyl-5-nitrophenyl)methanimine (7)

Yield: 82%, state = light-yellow solid, Melting Point = 178–179 °C; FTIR (KBr): (cm^{-1}), 3390 (NH), 3293 (Ar–CH), 2917 (sp^3-CH), 1604 (C = N), 1443 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6*): δ 11.41 (s, 1H, -NH), 9.20 (s, 1H, imine-H), 8.46 (s, 1H, Ar–H), 8.37 (s, 1H, Ar–H), 8.23 (d, 1H, J = 7.3 Hz, Ar–H), 8.05 (s, 1H, Ar–H), 7.95 (s, 1H, Ar–H), 7.86 (d, 1H, J = 7.2 Hz, Ar–H), 2.16 (s, 3H, Ar-CH_3); ^{13}C-NMR (150 MHz, DMSO-*d6*): δ 174.3, 162.3, 160.4, 150.4, 147.2, 147.1, 146.4, 144.5, 143.4, 140.2, 137.8, 128.7, 125.4, 124.6, 122.6, 115.2, 26.5; HREI-MS: m/z calcld for $C_{17}H_{11}ClN_6O_2S$, $[M]^+$ 398.1564 Found: 398.1534.

(E)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)−1-(p-tolyl)methanimine (8)

Yield: 58%, state = white solid, Melting Point = 147–148 °C; FTIR (KBr): (cm^{-1}), 3443 (NH), 3231 (Ar–CH), 2903 (sp^3-CH), 1621 (C = N), 1431 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6*): δ 11.51 (s, 1H, -NH), 9.26 (s, 1H, imine-H), 8.47 (d, 1H, J = 7.2 Hz, Ar–H), 8.37 (s, 1H, Ar–H), 8.31 (d, 2H, J = 8.4 Hz, Ar–H), 7.86 (d, 1H, J = 7.8 Hz, Ar–H), 7.62 (d, 2H, J = 7.8 Hz, Ar–H), 2.32 (s, 3H, Ar-CH_3); ^{13}C-NMR (150 MHz, DMSO-*d6*): δ 173.3, 163.5, 160.0, 149.6, 147.4, 146.2, 146.2, 142.2, 139.2, 137.4, 132.1, 123.6, 120.5, 119.4, 111.5, 30.9; HREI-MS: m/z calcld for $C_{17}H_{12}ClN_5S$, $[M]^+$ 353.1178 Found: 353.1157.

(E)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)−1-(4-fluorophenyl)methanimine (9)

Yield: 58%, state = green solid, Melting Point = 163–164 °C; FTIR (KBr): (cm^{-1}), 3442 (NH), 3243 (Ar–CH), 1622 (C = N), 1420 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6*): δ 11.48 (s, 1H, -NH), 9.19 (s, 1H, imine-H), 8.64 (d, 1H, J = 7.8 Hz, Ar–H), 8.52 (s, 1H, Ar–H), 7.76 (d, 2H, J = 8.0 Hz, Ar–H), 7.41 (d, 1H, J = 7.0 Hz, Ar–H), 7.05 (d, 2H, J = 7.2 Hz, Ar–H); 13C-NMR (150 MHz, DMSO-*d6*): δ 174.9, 167.5, 147.0, 144.8, 143.6, 142.2, 140.1, 139.4, 138.7, 137.9, 133.6, 126.6, 110.2, 107.3, 103.2; HREI-MS: m/z calcld for $C_{16}H_9ClFN_5S$, $[M]^+$ 357.2608 Found: 357.2517.

(E)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)−1-(4-chlorophenyl)methanimine (10)

Yield: 71%, state = Yellow solid, Melting Point = 189–190 °C; FTIR (KBr): (cm^{-1}), 3414 (NH), 3224 (Ar–CH), 1620 (C = N), 1398 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6*): δ 11.52 (s, 1H, -NH), 8.99 (s, 1H, imine-H), 8.40 (d, 1H, J = 7.5 Hz, Ar–H), 8.29 (d, 2H, J = 8.3 Hz, Ar–H), 8.21 (s, 1H, Ar–H), 8.03 (d, 1H, J = 7.9 Hz, Ar–H), 7.80 (d, 2H, J = 7.8 Hz, Ar–H); 13C-NMR (150 MHz, DMSO-d6): δ 175.8, 164.7, 150.9, 148.3, 145.9, 142.3, 142.2, 139.9, 136.1, 133.2, 128.9, 124.8, 123.4, 120.9, 120.8, 117.8, 117.3, 104.9; HREI-MS: m/z calcld for $C_{16}H_9C_{l2}N_5S$, $[M]^+$ 374.3374 Found: 374.3354.

(E)−1-(3-bromo-5-chlorophenyl)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)methanimine (11)

Yield: 74%, state = brown solid, Melting Point = 211–212 °C; FTIR (KBr): (cm^{-1}), 3318 (NH), 3243 (Ar–CH), 1613 (C = N), 1426 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6*): δ 11.62 (s, 1H, -NH), 9.16 (s, 1H, imine-H), 8.41 (s, 1H, Ar–H), 8.35 (s, 1H, Ar–H), 8.30 (d, 1H, J = 7.0 Hz, Ar–H), 8.11 (s, 1H, Ar–H), 8.06 (s, 1H, Ar–H), 7.80 (d, 1H, J = 7.3 Hz, Ar–H); ^{13}C-NMR (150 MHz, DMSO-*d6*): δ 172.9, 163.7, 160.7, 152.9, 147.8, 147.2, 146.1, 144.6, 143.4, 140.2, 137.7, 128.6, 125.9, 121.9, 120.9, 117.5; HREI-MS: m/z calcld for $C_{16}H_8BrCl_2N_5S$, $[M]^+$ 453.9261 Found: 453.9244.

(E)−1-(3-bromo-4-chlorophenyl)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)methanimine (12)

Yield: 51%, state = brown solid, Melting Point = 228–229 °C; FTIR (KBr): (cm^{-1}), 3402 (NH), 3218 (Ar–CH), 2935 (sp^3-CH), 1626 (C = N), 1428 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6*): δ 11.54 (s, 1H, -NH), 8.97 (s, 1H, imine-H), 8.51 (d, 1H, J = 7.0 Hz, Ar–H), 8.30 (d, 1H, J = 6.9 Hz, Ar–H), 8.19 (s, 1H, Ar–H), 8.13 (d, 1H, J = 8.4 Hz, Ar–H),

7.89 (d, 1H, J = 7.0 Hz, Ar–H); ^{13}C-NMR (150 MHz, DMSO-*d6): δ* 170.8, 168.2, 150.0, 150.0, 149.3, 146.8, 145.9, 140.6, 140.2, 139.8, 136.9, 133.7, 130.9, 130.4, 125.8, 122.2, 122.0; HREI-MS: m/z calcld for $C_{16}H_8Cl_2BrN_5S$, $[M]^+$ 453.2480 Found: 453.2456.

(E)−1-(4-bromophenyl)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)methanimine (13)

Yield: 59%, state = brown solid, Melting Point = 219–220 °C; FTIR (KBr): (cm^{-1}), 3411 (NH), 3215 (Ar–CH), 1624 (C = N), 1390 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6): δ* 11.93 (s, 1H, -NH), 9.03 (s, 1H, imine-H), 8.42 (d, 1H, J = 7.5 Hz, Ar–H), 8.33 (d, 2H, J = 6.3 Hz, Ar–H), 8.29 (s, 1H, Ar–H), 8.13 (d, 1H, J = 7.0 Hz, Ar–H), 7.81 (d, 2H, J = 7.8 Hz, Ar–H); ^{13}C-NMR (150 MHz, DMSO-*d6): δ* 172.5, 169.3, 150.9, 148.2, 146.4, 142.8, 142.7, 139.1, 136.8, 133.1, 130.9, 127.4, 123.0, 120.4, 120.4, 117.3, 117.3, 112.7; HREI-MS: m/z calcld for $C_{16}H_9BrClN_5S$, $[M]^+$ 418.5354 Found: 418.5323.

(E)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)−1-(3,4-dimethylphenyl)methanimine (14)

Yield: 59%, state = white solid, Melting Point = 147–150 °C; FTIR (KBr): (cm^{-1}), 3407 (NH), 3234 (Ar–CH), 2921 (sp^3-CH), 1628 (C = N), 1414 (C = C); ^{1}H-NMR (600 MHz, DMSO-*d6*): δ 12.39 (s, 1H, -NH), 8.92 (s, 1H, imine-H), 8.57 (d, 1H, J = 7.4 Hz, Ar–H), 8.36 (d, 1H, J = 6.5 Hz, Ar–H), 8.29 (s, 1H, Ar–H), 8.23 (d, 1H, J = 8.9 Hz, Ar–H), 7.94 (d, 1H, J = 7.0 Hz, Ar–H), 2.04 (s, 6H, Ar-CH_3); ^{13}C-NMR (150 MHz, DMSO-*d6): δ* 173.8, 163.7, 150.4, 149.0, 145.3, 144.8, 142.5, 140.4, 140.1, 139.8, 136.3, 134.7, 131.2, 130.7, 128.8, 124.8, 122.4, 34.5, 31.4; HREI-MS: m/z calcld for $C_{18}H_{14}ClN_5S$, $[M]^+$ 367.1269 Found: 367.1252.

(E)-N-(5-(3-chloro-1H-indazol-6-yl)−1,3,4-thiadiazol-2-yl)−1-(2,4-dimethylphenyl)methanimine (15)

Yield: 65%, state = white solid, Melting Point = 152–153 °C; FTIR (KBr): (cm^{-1}), 3452 (NH), 3245 (Ar–CH), 2951 (sp^3-CH), 1621 (C = N), 1432 (C = C);^{1}H-NMR (600 MHz, DMSO-*d6*): δ 11.65 (s, 1H, -NH), 8.89 (s, 1H, imine-H), 8.61 (d, 1H, J = 6.2 Hz, Ar–H), 8.42 (d, 1H, J = 6.0 Hz, Ar–H), 8.36 (s, 1H, Ar–H), 8.33 (d, 1H, J = 8.0 Hz, Ar–H), 8.04 (d, 1H, J = 7.4 Hz, Ar–H), 2.15 (s, 6H, Ar-CH_3); ^{13}C-NMR (150 MHz, DMSO-*d6): δ* 175.9, 171.2, 165.9, 149.5, 146.8, 144.2, 143.3, 140.5, 140.2, 139.0, 137.4, 134.7, 133.1, 130.2, 129.8, 125.4, 123.1, 31.8, 29.1; HREI-MS: m/z calcld for $C_{18}H_{14}ClN_5S$, $[M]^+$ 367.1432 Found: 367.1413.

Molecular docking protocol

Discovery studio visualizer (DSV), MGL tool 1.5.7 and auto dock vina were employed for the conduction of molecular docking study (Hussain et al. 2024a; Khan et al. 2024d, 2024a). The aim of this study is to analyze and investigate the novel synthesized derivatives **(1–15)** against the targeted enzymes (Hussain et al. 2024b; Khan et al. 2024c). Protein data bank (PDB) was used to recover the structure of enzymes. The first step is preparation of protein utilizing DSV in which already present ligand and water molecules were removed and saving of both prepared ligand and targeted protein in PDB format (Khan et al. 2024b; Zahoor et al. 2024). The next step was carried out in auto dock in which polar hydrogen and gasteiger charges were added to protein. In the last step DSV was used for the opening of dock protein and ligand and for the identification of binding interaction of ligand with enzyme (Ullah et al. 2024; Khan et al. 2023a, b).

Pharmacokinetic parameters

Additionally, ADMET properties of newly designed compounds were predicted through SwissADME (http://www.swissadme.ch) which was utilized to evaluate the synthetic accessibility of the newly developed compounds. Computer-aided ADME studies were performed using webserver pkCSM (http://biosig.unimelb.edu.au/pkcsm/prediction) to estimate the pharmacokinetic features of the prepared inhouse compounds.

Assay protocol of thymidine phosphorylase

In accordance with the previously reported method, assay protocol has been adopted.

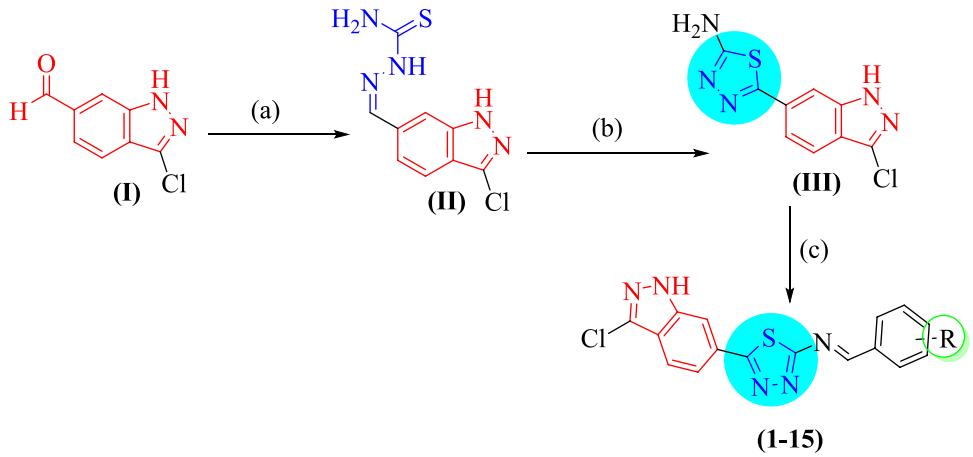
Assay protocol of α-glucosidase

In accordance with the previously reported method, assay protocol has been adopted.

Results

Chemistry

This study aims to synthesize novel 3-chloro-1*H*-indazole based 1,3,4-thiadiazole derivatives (**1–15**). In the first step, 3-chloro-1*H*-indazole-5-carbaldehyde **(I)** and thiosemicarbazide are mixed in ethanol and catalytic amount of CH_3COOH and refluxed. This mixture results in 3-chloro-1*H*-indazole-based thiosemicarbazone intermediate **(II)**,



Scheme-1 Protocol for the reaction condition of 3-chloro-1*H*-indazole based 1,3,4-thiadiazole derivatives (1–15) synthesis: **a)** EtOH/AcOH, thiosemicarbazide, Reflux 6 h; **b)** 1,4-dioxane, I_2 (Molecular Iodine), K_2CO_3, Reflux 13–14 h; **c)** EtOH, Et_3N, variously substituted Aldehyde and reflux for 6 h

after stirring for 6 h. Addition of molecular iodine leads to oxidative cyclization followed by addition of potassium carbonate in 1,4-dioxane drop wisely in reaction mixture and stirring for 13–14 h. TLC were conducted to measure the progress of the reaction after every hour. When the reaction was completed, the solvent was evaporated using reduced pressure resulting in the formation of 3-chloro-1*H*-indazole-based 2-amino-thidaizole substrate **(III).** This **(III)** was dissolved in ethanol, following addition of triethylamine as base and differently substituted aldehydes. Until the complete conversion into 3-chloro-1*H*-indazole based 1,3,4-thiadiazole derivatives (**1–15**) in appreciable yields the reaction mixture was kept stirring. The structural characteristics of 3-chloro-1*H*-indazole based 1,3,4-thiadiazole derivatives (**1–15**) were investigated using FTIR, NMR (^{1}H-NMR and ^{13}C-NMR) and HREI-MS were shown in Scheme 1. The substitution pattern of the synthesized compounds along with their inhibitory potential against thymidine phosphorylase and α-glucosidase enzymes were mentioned in Table 1.

Discussion

Biological profiles of the synthesized 3-chloro-1H-indazole based 1,3,4-thiadiazole derivatives (1–15)

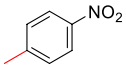
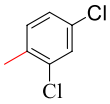
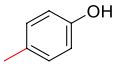
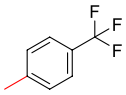
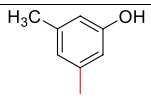
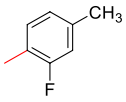
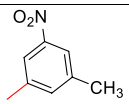
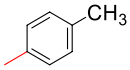
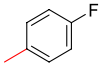
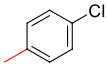
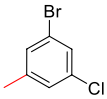
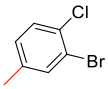
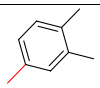
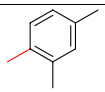
Thymidine phosphorylase enzyme inhibitory activity

The inhibition potential of 3-chloro-1*H*-indazole based 1,3,4-thiadiazole derivatives (**1–15**) against thymidine phosphorylase enzyme was compared to standard drug candidates 7 Deazaxanthine, 7DX (IC_{50} = 12.42 ± 1.27 μM). The nature and position of substituents plays a significant role in inhibitory potential of derivatives. The main reasons is due to varied interaction with the targeted enzyme with the active sites of the synthesized molecules. The inhibition potency of most of the derivative was moderate to good within the range of IC_{50} = 4.70 ± 1.34 to 19.23 ± 0.28 μM. The excellent biological activity was shown by the derivatives having -F, -Cl, -CF_3, -OH, and NO_2 substituents on the phenyl ring. The presence of substituents on para-position enhanced the inhibitory potential whereas the presence of bulky group caused a decreased inhibitory potential as compared to the standard drug. This can be attributed to reducing interaction with targeted enzymes. Substitution behavior and effective inhibition of both enzymes thymidine phosphorylase and α-glucosidase are explained in the structure activity relationship in detail along with their IC_{50} values mentioned in Table 1.

Structure activity relationship (SAR) of thymidine phosphorylase

The standard drug candidates 7 Deazaxanthine, 7DX (IC_{50} = 12.42 ± 1.27 μM), was taken as reference for the comparison of the potency of newly synthesized 3-chloro-1*H*-indazole based 1,3,4-thiadiazole derivatives (**1–15**). Firstly hydroxyl (-OH) group is present in both the analog **3** and **5** at various positions i.e., para and meta, respectively. The derivative **3** with hydroxy at para position has shown better activity (IC_{50} = 7.92 ± 0.73 μM), and the derivative **5** with hydroxy at meta position has shown poor activity (IC_{50} = 9.10 ± 0.74 μM). Another reason for the poor inhibition potential of derivative **5** can be attributed to the presence of methyl group on the meta position of the phenyl ring which increases the steric hindrance. Fluoro substituted derivatives have also demonstrated varied inhibition potential for derivative **4** (IC_{50} = 4.70 ± 1.34 μ M) and **9** (IC_{50} = 5.05 ± 0.19 μM). Similarly, the derivatives with substitution of nitro group at para position (**1**) have shown better efficacy than meta NO_2 substitution (**7**). The para substitution enhances the hydrogen bonding with the targeted molecule. The lower inhibitory activity of derivative (**7**) can also be due to presence of methyl group which causes steric hindrance. In derivative (**11, 12, 13, 14**) and (**15**) the presence of bulky group (Br)

Table 1 Representation of the attached substituents with phenyl ring along with Enzymes inhibitory profiles of each molecule against thymidine phosphorylase and α-glucosidase enzymes

S/No	R	Thymidine Phosphorylase IC_{50} (μM ± SEM)	α-glucosidase IC_{50} (μM ± SEM)
1		7.70 ± 0.29	3.81 ± 0.63
2		6.94 ± 0.56	4.98 ± 1.09
3		7.92 ± 0.73	4.70 ± 0.30
4		4.70 ± 1.34	1.38 ± 0.46
5		9.10 ± 0.74	6.32 ± 0.87
6		7.82 ± 0.27	5.35 ± 0.29
7		8.29 ± 0.52	6.31 ± 0.77
8		15.33 ± 1.59	9.59 ± 0.11
9		5.05 ± 0.19	2.40 ± 0.64
10		7.62 ± 0.36	5.27 ± 1.21
11		17.30 ± 0.35	12.53 ± 1.33
12		17.74 ± 0.44	13.28 ± 0.51
13		19.23 ± 0.28	16.95 ± 0.79
14		16.98 ± 0.24	10.29 ± 0.18
15		17.52 ± 0.83	13.28 ± 0.71
Standard drugs		**7-Deazaxanthine, 7DX (IC_{50} = 12.42 ± 1.27 µM)**	**Acarbose (IC_{50} = 5.97 ± 0.35 µM)**

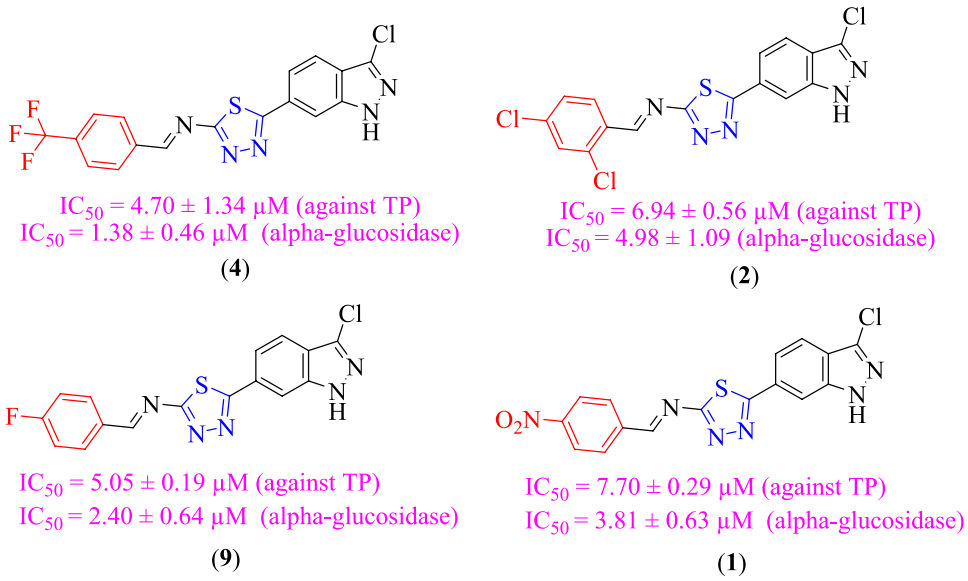


Fig. 2 Highly promising active analogues such as compound **4, 2, 9** and **1**

and CH_3 weakens the interaction with the enzymes leading to poor inhibition potential. The presence of -Cl group at para position of the phenyl ring in derivative **2** and **10** has significant biological potency due to smaller size and higher electronegativity of -Cl. In the derivatives **4, 9, 2, 1** and **6**, the position of substituent is same which leads to similar inhibition potential of these derivatives against thymidine phosphorylase enzyme along with their IC_{50} values mentioned in Table 1.

Structure activity relationship (SAR) of α-glucosidase

All the novel synthesized derivatives 1,3,4-thiadiazole derivatives (**1–15**) were investigated against α-glucosidase enzyme. The inhibition potential shown by the derivatives is comparable to the standard drug acarbose (IC_{50} = 5.97 ±0.35 µM). The variety in the substitution leads to a wide range of inhibition potential against α-glucosidase enzyme. The same procedure is adopted to investigate better binding

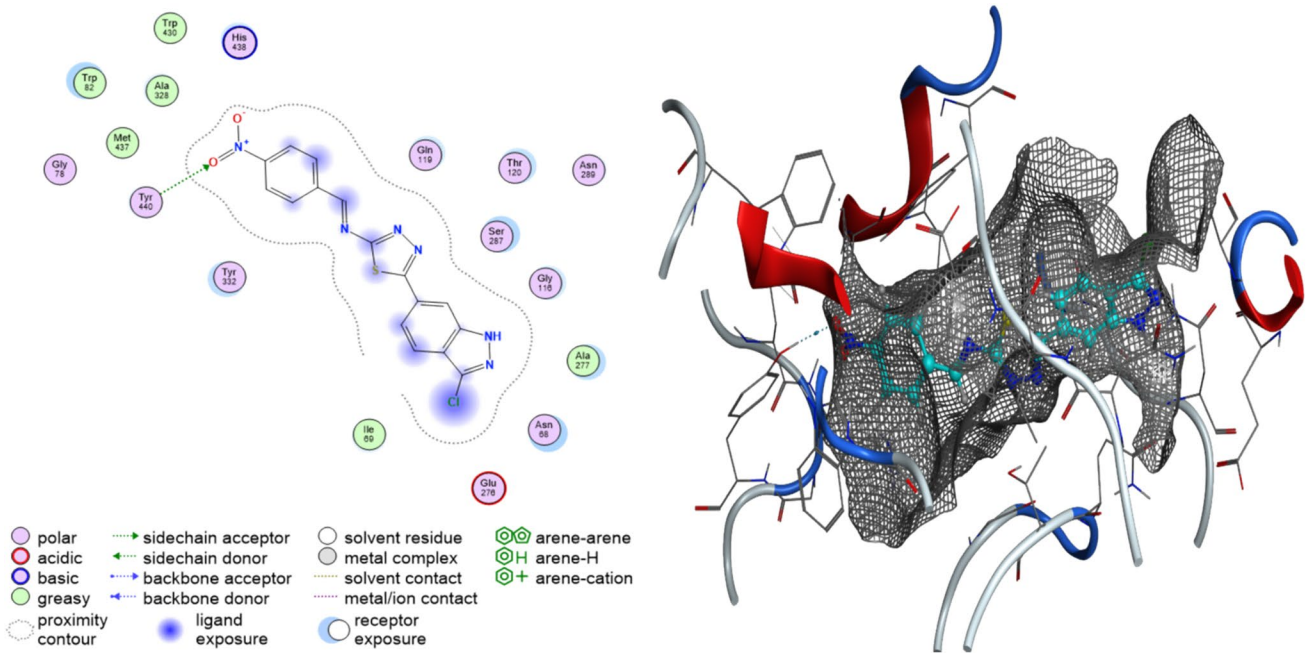


Fig. 3 Binding interactions of analog 1 with α-glucosidase

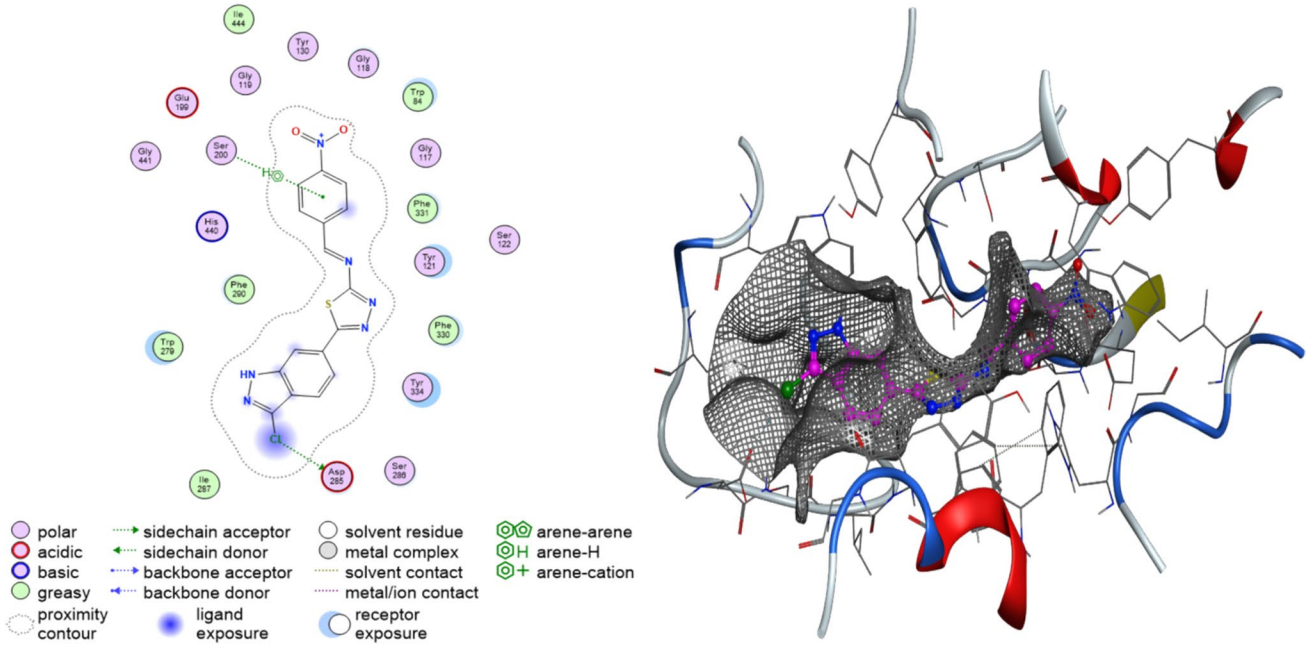


Fig. 4 Binding interactions of analog 1 with thymidine phosphorylase

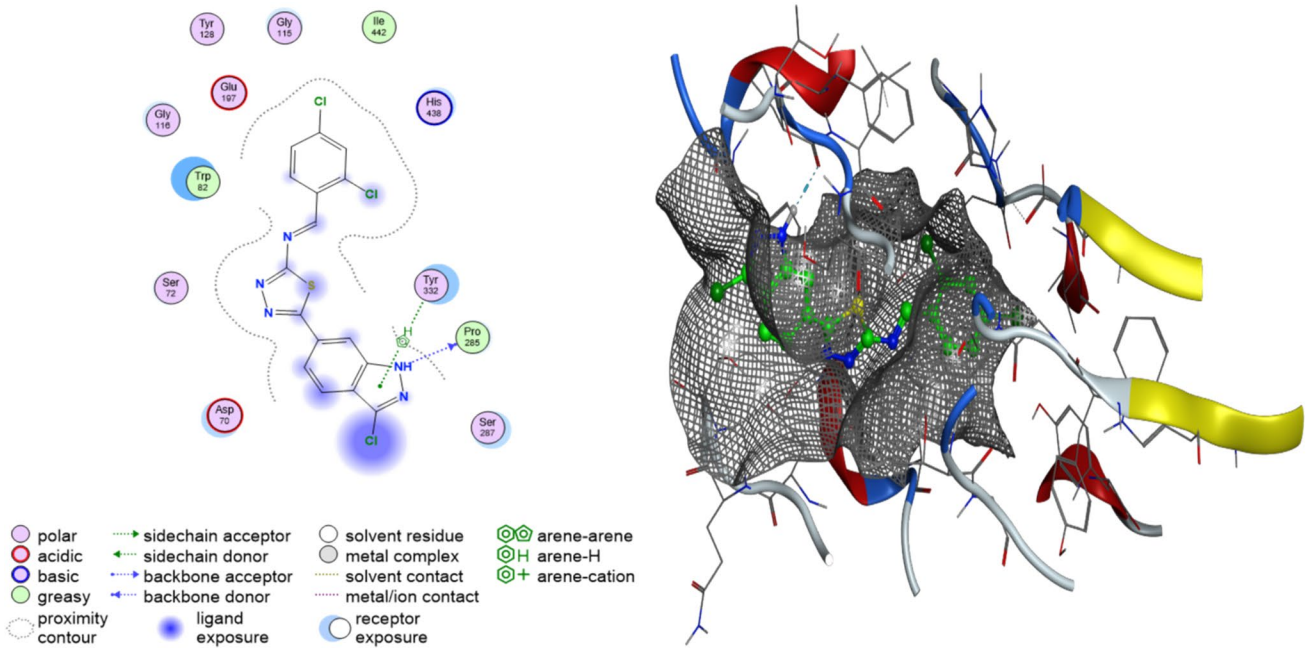


Fig. 5 Binding interactions of analog 2 with α-glucosidase

modalities and effective agent. The variation in the position of the same substituents either enhances or decreases the inhibitory potency of the derivatives. Derivative **1** (IC_{50} = 1.38 ± 0.46 μM), **9** (IC_{50} = 2.40 ± 0.64 μM), **1** (IC_{50} = 3.81 ± 0.63 μM), **2** (IC_{50} = 4.98 ± 1.09 μM) and **3** (IC_{50} = 4.70 ± 0.30 μM) has shown the effective biological activity. This could be attributed to the presence of *para*-substitution on the ring which forms stable interactions with the α-glucosidase enzyme. The derivatives containing -CF_3, -F, -OH, -NO_2, and -Cl as substituents have shown the best potency. Among these substituents, the derivative having maximum number of -F groups have shown the maximum inhibition potential against targeted enzyme. The derivatives with fewer number of activation groups have shown lower

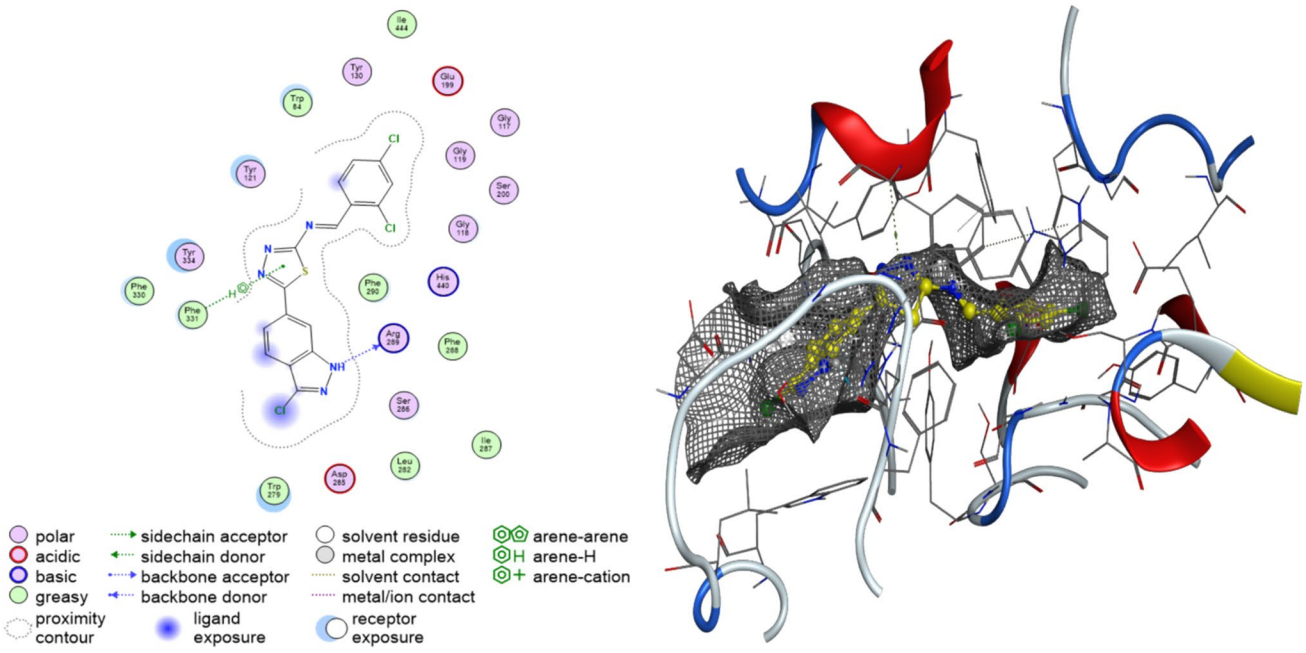


Fig. 6 Binding interactions of analog 2 with thymidine phosphorylase

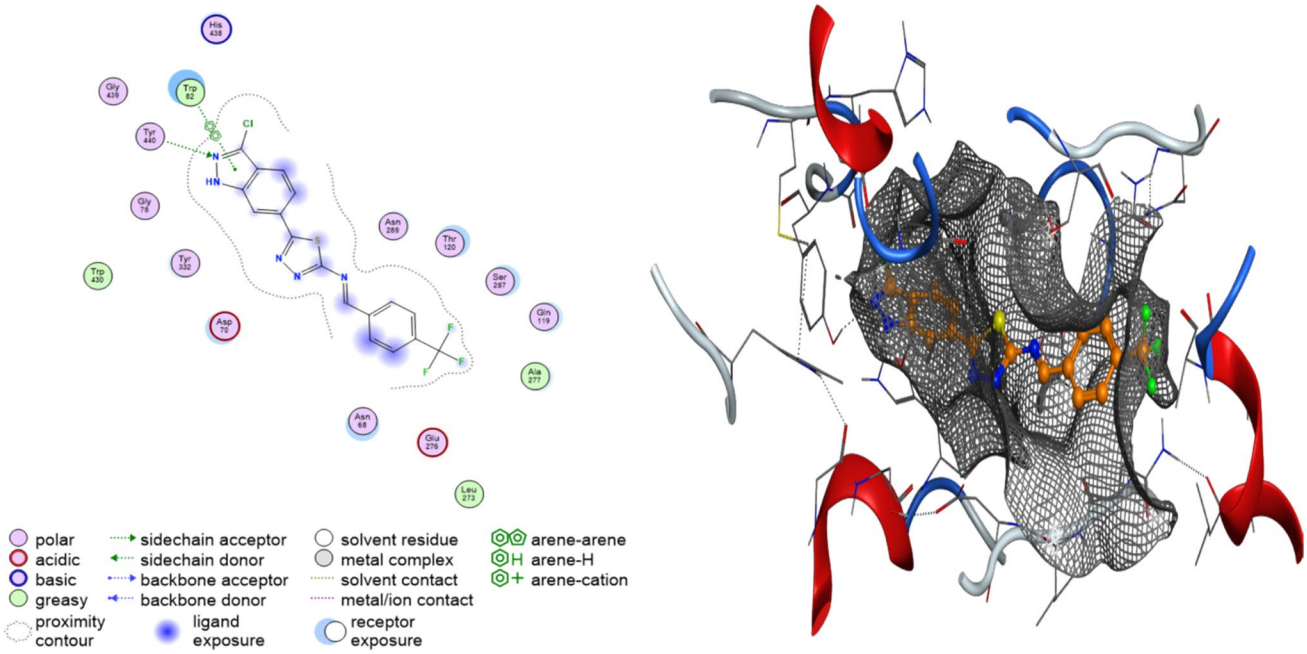


Fig. 7 Binding interactions of analog 4 with α-glucosidase

efficacy due to lower interaction were shown in Fig. 2 along with their IC_{50} values mentioned in Table 1.

The derivatives having -CH_3 group has shown moderate to poor inhibition potency due to steric hindrance. In the same manner the presence of bulky groups like -Br leads to decreased efficacy. The presence of substituents is very decisive in determining the biological potency of derivatives. The ring de-activating groups, or the bulky groups decreases the interaction with the enzyme leading to smaller efficacy. Few derivatives have shown the best inhibition potential as compared to standard drug acarbose. The derivatives which have shown better inhibition potential are further investigated in molecular docking studies to evaluate the binding interactions of ligand with targeted enzyme.

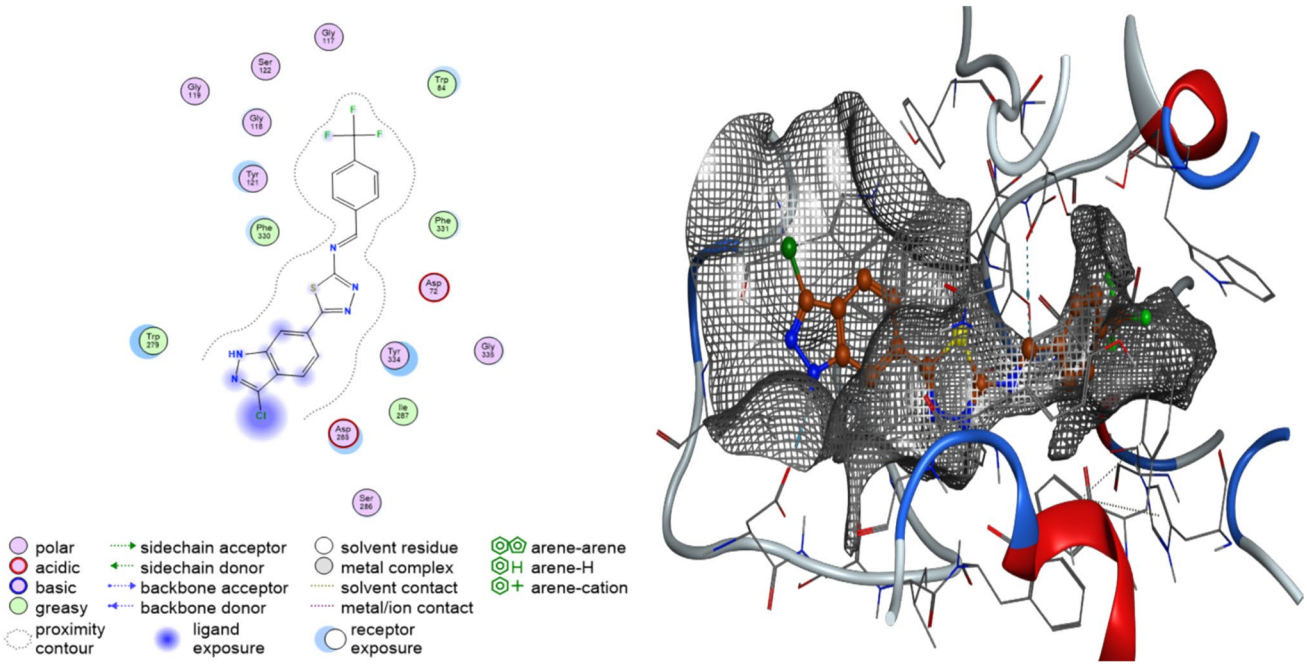


Fig. 8 Binding interactions of analog 4 with thymidine phosphorylase

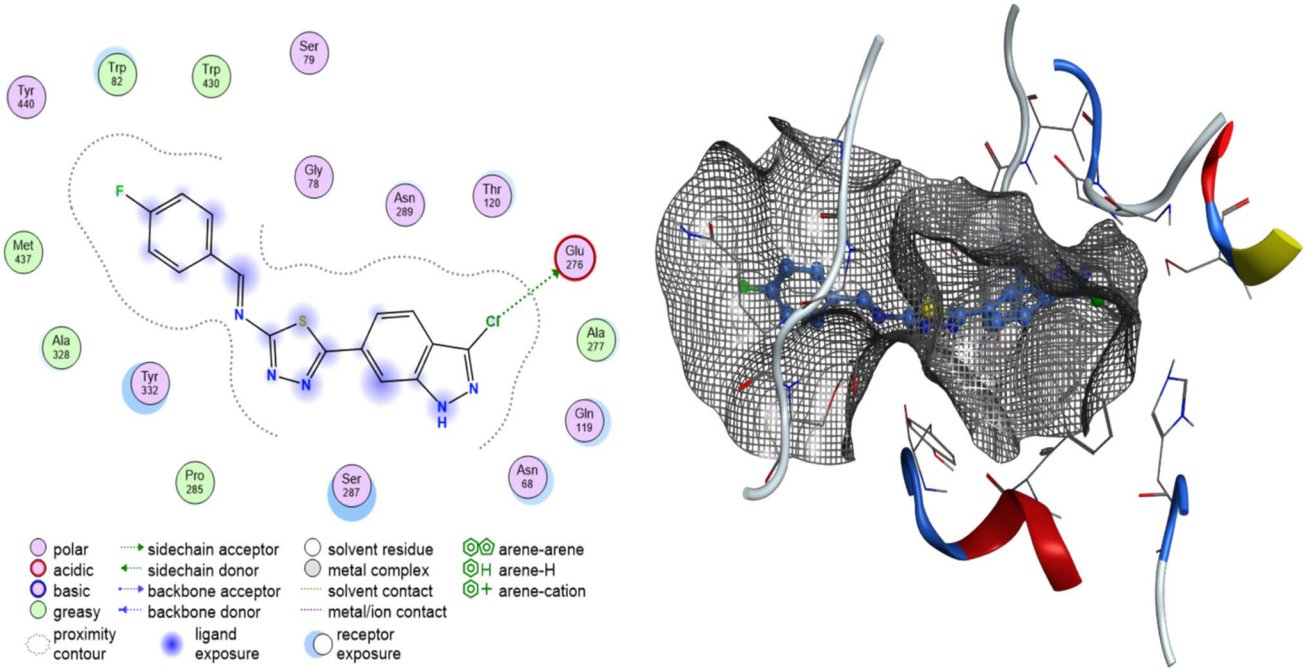


Fig. 9 Binding interactions of analog 9 with α-glucosidase

Molecular docking

Molecular docking study is an emerging in silico investigatory tool to analyze different mode of interactions and various interactive properties shown by ligands against different enzymes. This in silico study has significant importance in determining the biological inhibition of potent molecule. We have also conducted molecular docking investigation of the potent compounds from various number series which were found as the leading inhibitors in biological assessment

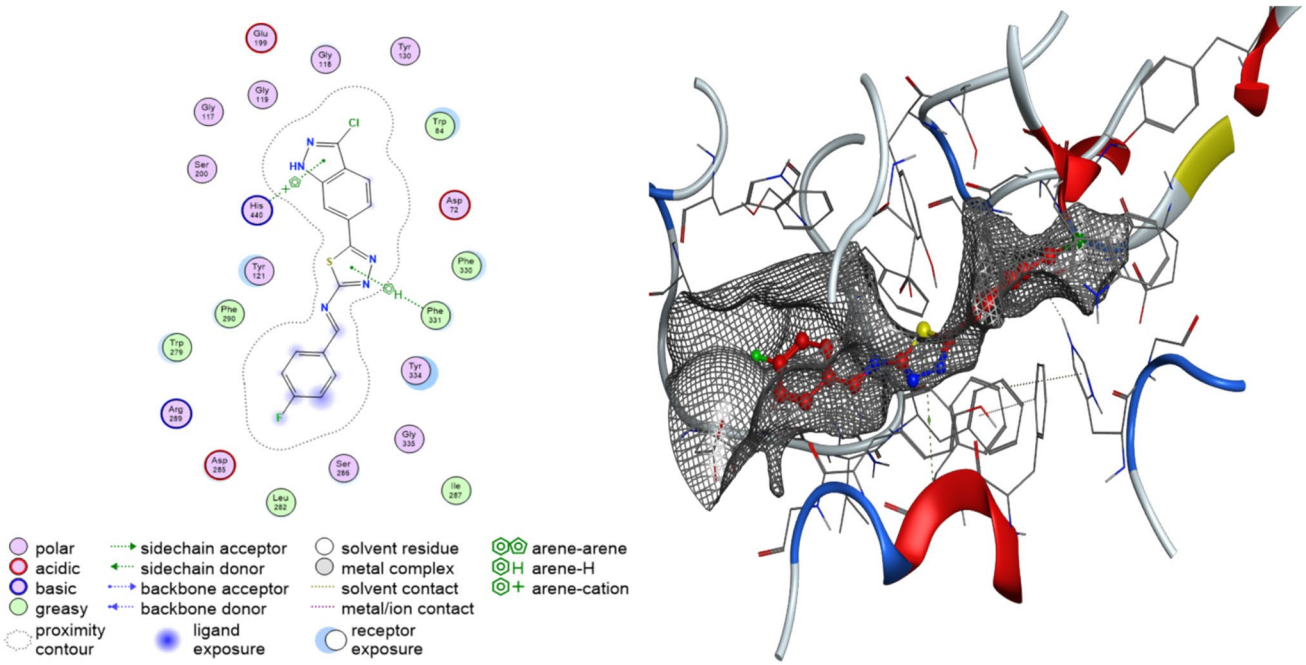


Fig. 10 Binding interactions of analog 9 with thymidine phosphorylase

(Khan et al. 2022). For this purpose, different docking tools including Auto dock Vina, pymol and discovery studio visualizer as well as we retrieved proteins from the online source RCSB protein data bank (PDB) using their specific identifier codes. The specific code for α-glucosidase is 3N37and thymidine phosphorylase is 1UOU. Protein complex was optimized, and the ligands were docked in this protein complex. Versatile possess were observed in docking, and the geometry having minimum docking score and highest binding affinity were identified and visualized by docking tools. This visualization and complete procedure give insight into the different binding interactions of potent molecules with targeted enzymes. These ligands have various significant heteroatoms and attached functionalities, and due to this they showed maximum binding interactions. They have a minimum docking score having a range of −12 to −14, and, they have RMSD range of (1.2–1.5). They caused different effective changes in interactions and orientation of the protein complex, in which these ligands were docked, as they make new strong and effective binding interactions. Mostly they contain polar and acidic species, and they take part in different interactions. They have the ability to make side chain donors as well as acceptor and also their electronic density region are much expose to enzyme for making strong binding folds. All these interactions were illustrated in Figs. 3, 4, 5, 6, 7, 8, 9 and 10.

Pharmacokinetics analysis

Among the tested compounds analyzed, Compound-**9** emerges as the most promising drug-like candidate. It demonstrates a highly favorable pharmacokinetic profile, with high GI absorption, no violations of key drug-likeness rules (such as Lipinski's, Veber, and Egan), and optimal permeability (log Kp = −5.39). These characteristics suggest that Compound-**9** is well-suited for oral administration, with a good balance between lipophilicity and solubility (XLOGP3 = 5.04), which are crucial for its absorption and bioavailability. Its lack of PAINS and Brenk alerts further indicates that it is unlikely to possess toxic or undesirable structural motifs, making it a strong candidate for further development. In comparison to 7-Deazaxanthine, which also exhibits a similar favorable pharmacokinetic profile, Compound-**9** holds significant promise as a lead compound for drug development.

On the other hand, Compound-**2** and Compound-**4** share some similarities in terms of high lipophilicity and good GI absorption, but they both exhibit concerning issues with CYP enzyme inhibition, including interactions with CYP1A2, CYP2C19, and CYP2C9. These interactions could pose risks for drug–drug interactions and metabolic toxicity, potentially limiting their use in therapeutic applications. Despite these concerns, these compounds are more

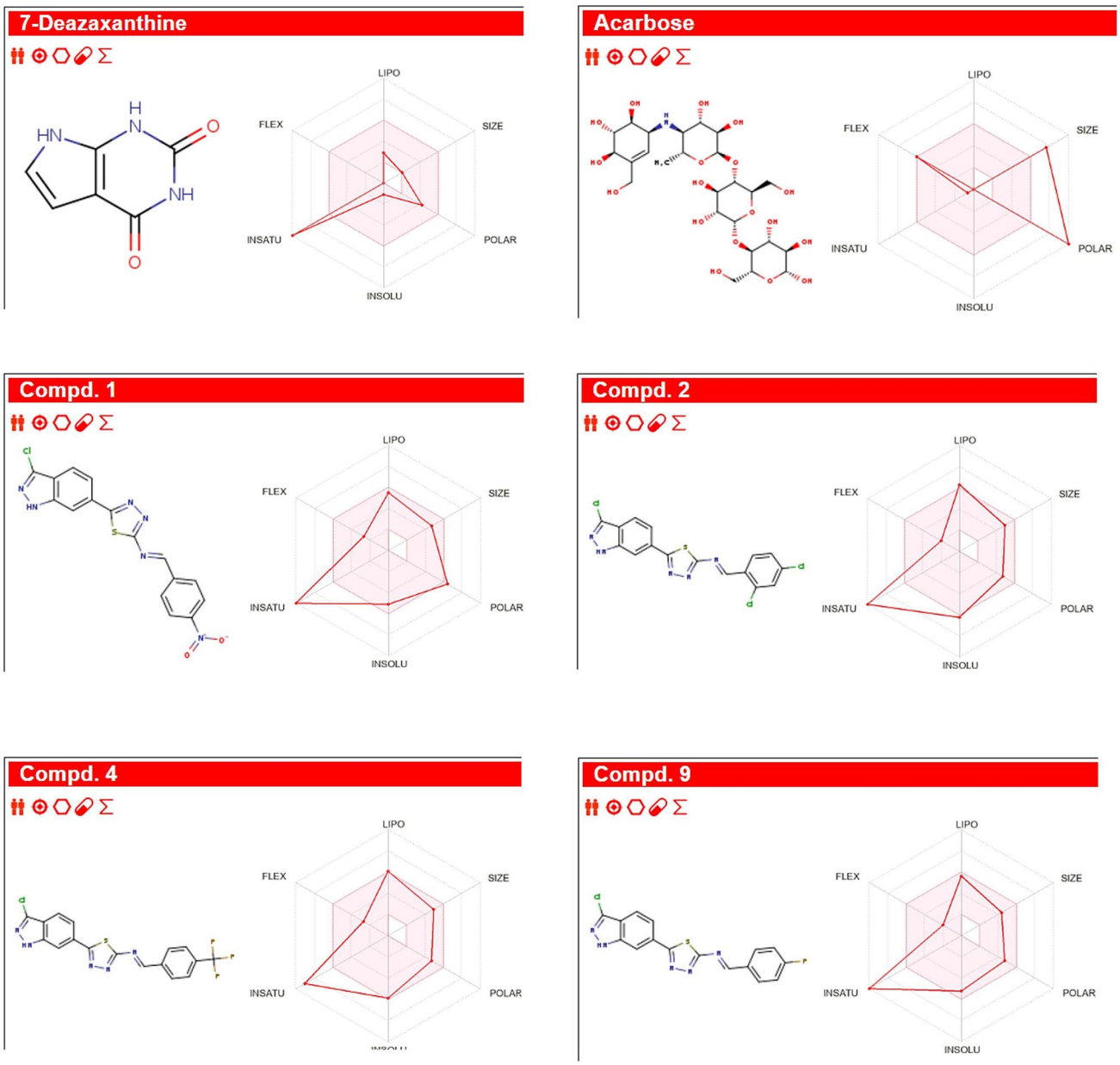


Fig. 11 Bioavailability radar of compounds (**1, 2, 4, 9**) using the SwissADME server

absorbable than Acarbose, which is a marketed drug but has its own limitations in terms of oral bioavailability and systemic absorption. Compound-**2** and Compound-**4** also violate several drug-likeness filters (such as Lipinski and Ghose), and while they show some promise in terms of absorption, their metabolic liabilities need to be addressed in order to improve their suitability for systemic use.

Compound-**1**, while moderately drug-like, has some notable drawbacks. It shows no Lipinski violations, but it suffers from poor solubility and low GI absorption, which could hinder its bioavailability and effectiveness as an oral drug. Additionally, the presence of Brenk alerts suggests the possibility of undesirable substructures, further complicating its development. While it is more drug-like than Acarbose, which is known for its poor systemic bioavailability, Compound-**1** still requires significant optimization in terms of solubility and absorption to reach a more favorable drug profile.

Overall, Compound-**9** stands out as the most promising candidate for drug development, offering a balance

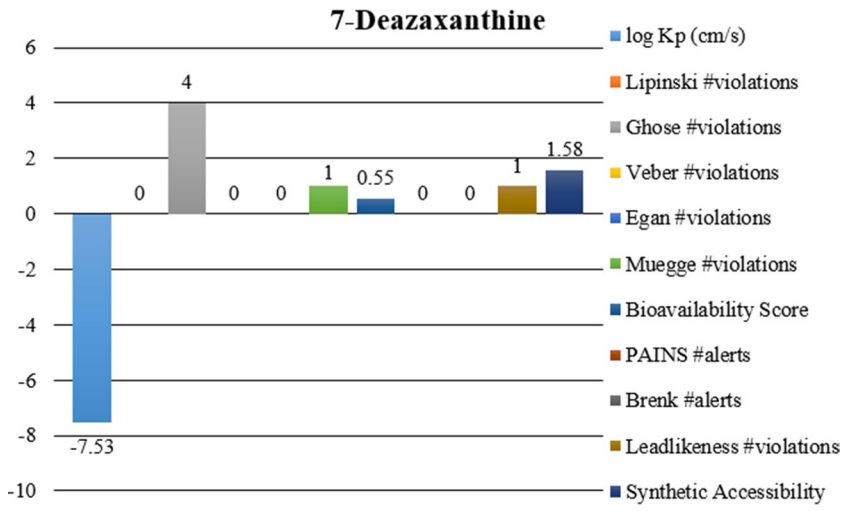


Fig. 12 ADME analysis of 7-Deazaxanthine

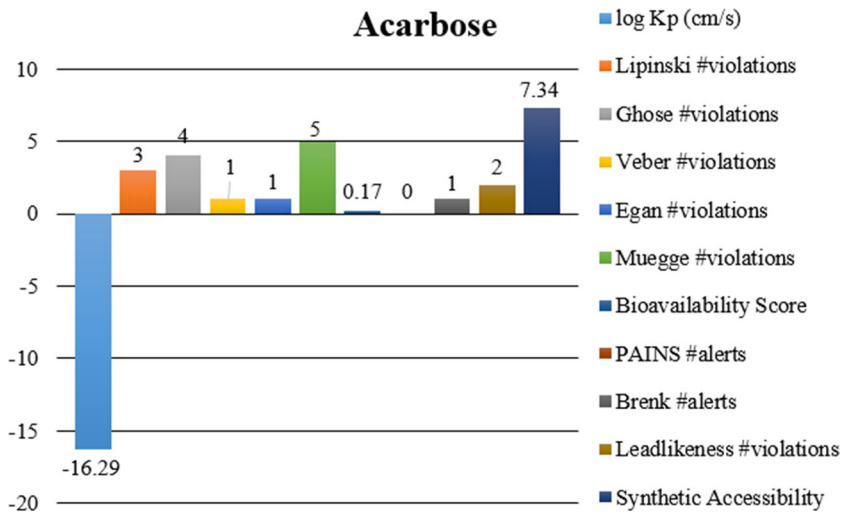


Fig. 13 ADME analysis of Acarbose

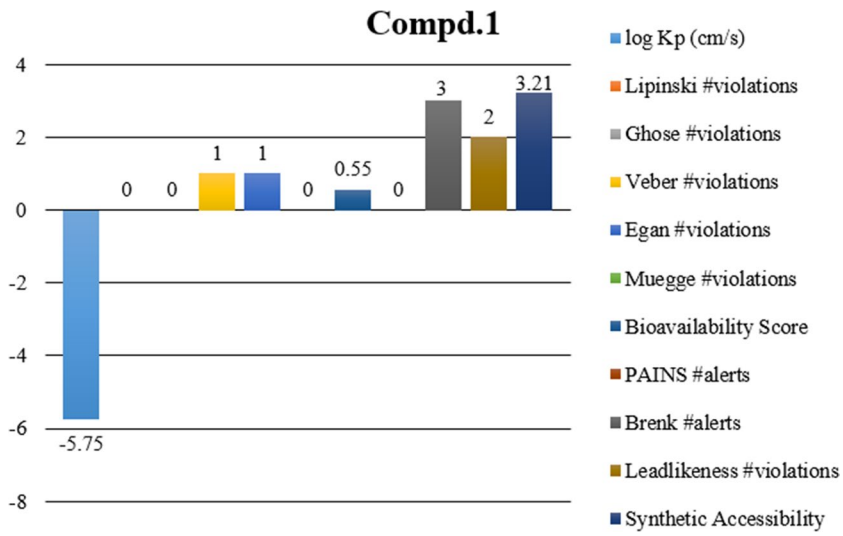


Fig. 14 ADME analysis of Compound−**1**

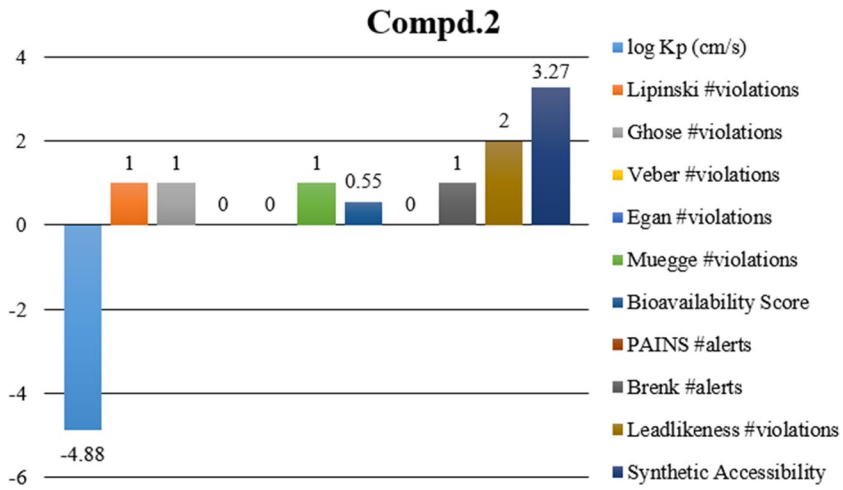


Fig. 15 ADME analysis of Compound—**2**

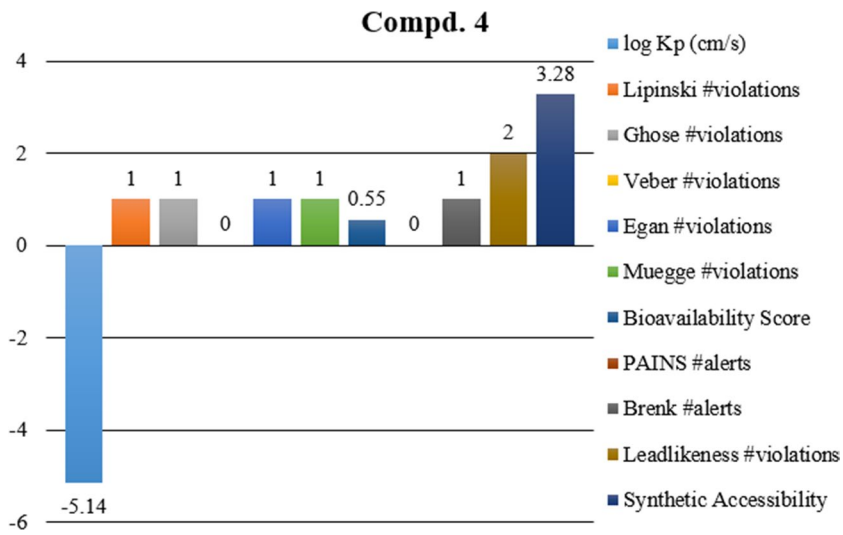


Fig. 16 ADME analysis of Compound—**4**

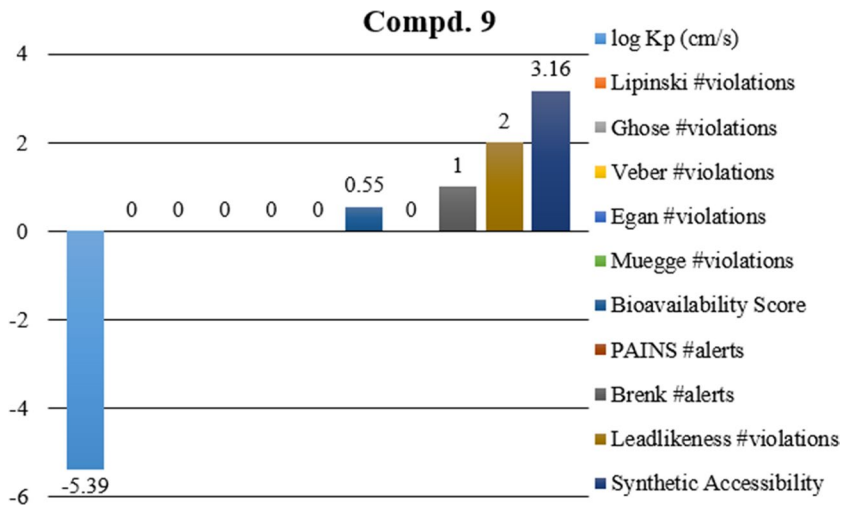


Fig. 17 ADME analysis of Compound—**9**

Table 2 Pro-Tox II toxicological parameters of the bioactive compounds **1, 2, 4, 9**

Toxicity	Compound-1	Compound-2	Compound-4	Compound- 9
Predicted LD50 (mg/kg)	1100	500	500	500
Predicted Toxicity Class (Class 1–3: toxic, lethal if consumed; Class 4–5: moderate toxicity, harmful if consumed, Class 6: not toxic)	4	4	4	4
Hepatotoxicity	Yes	Yes	Yes	Yes
Carcinogenicity	Yes	No	No	No
Immunotoxicity	No	No	No	No
Mutagenicity	Yes	Yes	Yes	Yes
Cytotoxicity	No	No	No	No
BBB permeability	Yes	Yes	Yes	Yes
Cytochrome CYP2D6	No	No	Yes	No
Cytochrome CYP3A4	No	Yes	Yes	Yes
Cytochrome CYP1A2	Yes	No	Yes	No
Cytochrome CYP2C19	No	No	No	No
Cytochrome CYP2C9	No	No	Yes	Yes

of favorable properties such as high absorption, permeability, and lack of rule violations, positioning it closer to 7-Deazaxanthine in terms of its drug-likeness. Compound-**2** and Compound 4 have potential but require modifications to address their lipophilicity and metabolic issues, while Compound-**1** needs improvements in solubility and absorption. Acarbose, despite being a marketed drug, serves as a benchmark of a molecule with poor bioavailability and limited systemic action, highlighting the room for improvement in these compounds. Therefore, Compound-**9** is the most suitable for further development, with the best potential for oral bioavailability and overall therapeutic success are presented in Figs. 11, 12, 13, 14, 15, 16, 17.

The Pro-Tox II analysis indicates that all selected compounds fall under toxicity class 4, suggesting moderate toxicity and potential harm if consumed, with a predicted LD_{50} of 500–1100 mg/kg. All compounds are predicted to be hepatotoxic and mutagenic, raising concerns about potential liver damage and genetic alterations. Compound**−1** is also flagged as carcinogenic, unlike the others. None of the compounds exhibit immunotoxicity or cytotoxicity, which is favorable. All compounds show blood–brain barrier (BBB) permeability, indicating potential effects on the central nervous system. Regarding cytochrome interactions, variation is observed: Compound**−4** interacts with the most CYP enzymes (CYP2D6, CYP3A4, CYP1A2, and CYP2C9), suggesting a higher risk of metabolic interactions. Overall, while the compounds are not highly toxic, their mutagenic, hepatotoxic, and BBB-penetrating nature warrants careful evaluation before therapeutic application are explained in Table 2 and Fig. 18.

The StopTox toxicity analysis of the selected compounds reveals moderate levels of toxicity across various parameters. Compound-**1** showed slightly higher toxicity in terms of eye irritation and skin irritation (75% and 80%, respectively), indicating a greater potential to cause irritation compared to the others. Compound**−4** exhibited the highest acute oral toxicity (72%), suggesting a higher risk if ingested. Overall, all compounds showed consistent moderate skin sensitization potential (60%) and similar acute dermal toxicity (ranging from 52 to 57%). While none of the compounds are classified as highly toxic, they all pose potential risks particularly for eye and skin exposure which should be carefully considered during formulation and handling are presented in Table 3.

Conclusion

This study presents the synthesis of novel 3-chloro-1*H*-indazole-based 1,3,4-thiadiazole derivatives (**1–15**), which were evaluated for their biological activity against thymidine phosphorylase and α-glucosidase enzymes. The results indicate that the biological activity of the derivatives varies significantly, with the nature of substituents on the phenyl ring playing a critical role in determining potency. Notably, compounds-**4** ($IC_{50} = 4.70 \pm 1.34$ μM and $IC_{50} = 1.38 \pm 0.46$ μM), **9** ($IC_{50} = 5.50 \pm 0.19$ μM and $IC_{50} = 2.40 \pm 0.64$ μM), **2** ($IC_{50} = 6.94 \pm 0.56$ μM and $IC_{50} = 4.98 \pm 1.09$ μM), **1** ($IC_{50} = 7.70 \pm 0.29$ μM and $IC_{50} = 3.81 \pm 0.63$ μM) and **6**

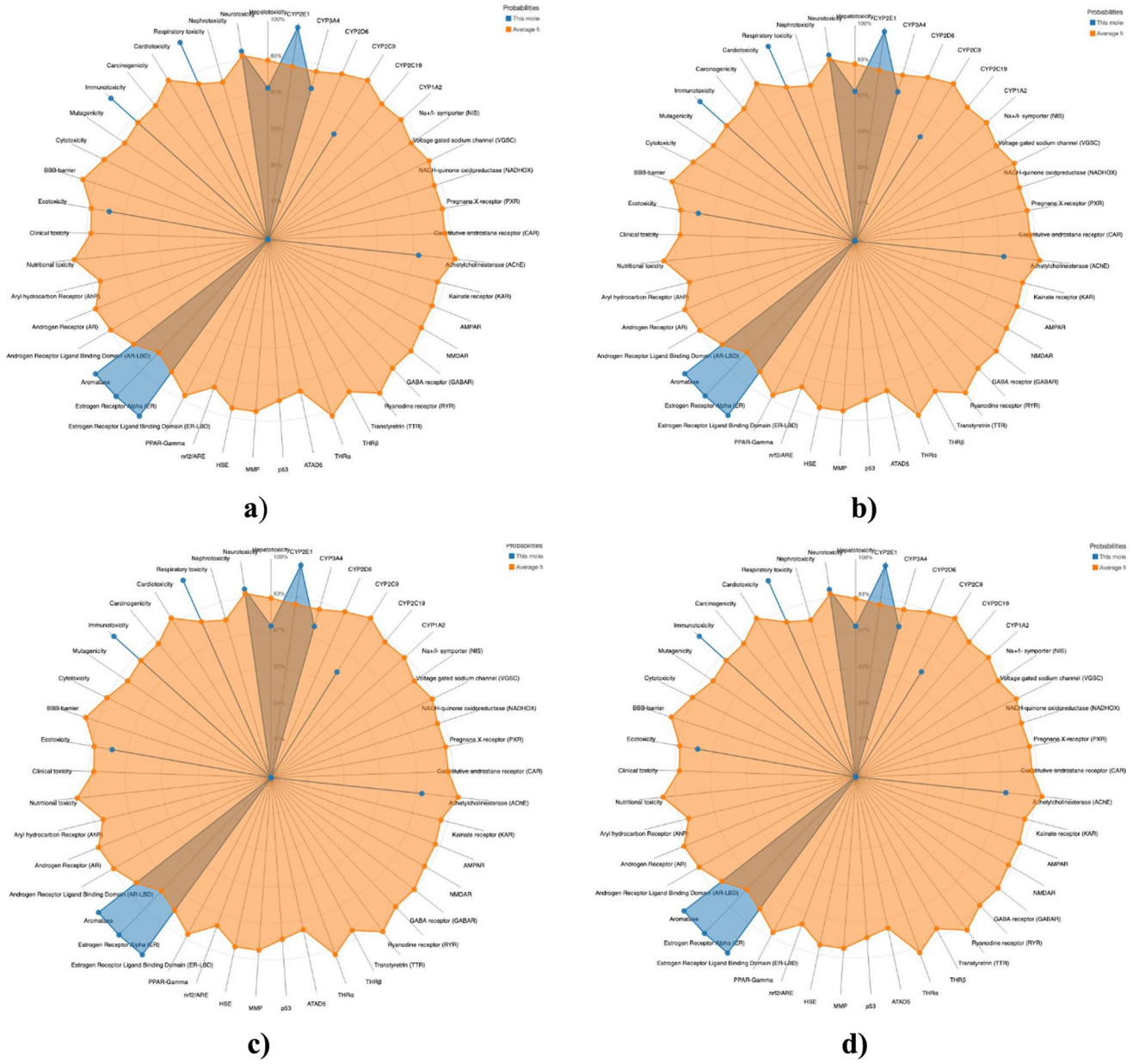


Fig. 18 The toxicity radar and network chart of Compound**−1 (a)**, Compound**−2 (b)**, Compound**−4 (c)**, Compound**−9(d)**

(IC_{50} = 7.82 ± 0.27 μM and IC_{50} = 5.35 ± 0.29 μM), exhibited strong inhibition against both thymidine phosphorylase and α-glucosidase enzymes, with IC_{50} values ranging from 4.70 ± 1.34 μM to 7.82 ± 0.27 μM. These derivatives showed promising activity compared to standard drugs, 7-Deazaxanthine (IC_{50} = 12.42 ± 1.27 μM) and acarbose (IC_{50} = 5.97

Table 3 StopTox toxicity parameters of the selected compounds **1, 2, 4** and **9**

Compound	Acute Inhalation Toxicity %	Acute Oral Toxicity	Acute Dermal Toxicity	Eye Irritation and Corrosion	Skin Sensitization	Skin Irritation and Corrosion
1	55	65	57	75	60	80
2	54	60	52	69	60	70
4	50	72	53	67	60	60
9	53	69	52	67	60	70

±0.35 μM). Moreover, molecular docking and ADMET studies further confirmed the favorable binding interactions and drug-like properties of these compounds, suggesting their potential for future therapeutic development.

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Author contributions Y. Khan; Project Supervisor, Methodology, data curation, Conceptualization. A. Maalik; original draft, Write-up, Experimental, H. Sarfraz; Visualization, Conceptualization. S. Khan; formal analysis, Software's, Rafaqat Hussain; Formal Analysis, Wajid Rehman; writing—review and editing, Marouan Kouki, Software, characterization, analyze data, Aisha Usman: formal analysis, Rubina Adnan; Software, analyze data, formal analysis. The authors declare that all data were generated in-house and that no paper mill was used.

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Data availability All source data for this work (or generated in this study) are available upon reasonable request.

Declarations

Ethics approval All synthesis experiments were complied with the guidelines and performed in accordance with the protocol of Chemistry Laboratory, Department of Chemistry, COMSATS University Islamabad, with the collaboration of Department of Chemistry, Hazara University Mansehra, Pakistan.

Consent to participate Written informed consent was obtained from each subject.

Consent for publication Written informed consent for publication was obtained from all participants.

Competing Interests The authors declare no competing interests.

Competing interests The authors declare no competing interests.

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